

VARUN PAL

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EDUCATION

University of Illinois Urbana-Champaign

Masters of Science in Computer Engineering

Bachelor of Science in Computer Engineering

- GPA: 3.93

- Courses: Comp Arch (ece411)

Data Structures and Algs (cs225)

Deep Learning for CV (cs444)

OS Design (ece391, cs423)

GPU Programming (ece408)

Machine Learning (cs446)

Distributed Systems (cs425)

Computer Networking (ece438)

Computer Systems Analysis (ece541)

Urbana-Champaign, IL

Expected Graduation: Fall 2027

Graduated: Spring 2026

WORK EXPERIENCE

Cisco Systems

May 2026 – Aug 2026

Software Engineering Intern

San Jose, CA

- Re-architected optic dashboard's MongoDB backend, cutting query latency from 16 min to 100 ms (9,600x)
- Replaced $O(n)$ scans over 8.4M raw sensor records with $O(1)$ reads of daily pre-aggregated snapshots; cut storage from $O(n^2)$ to $O(n)$ in 95% of cases (worst case: daily optic swaps), so storage savings grow over time
- Built peer optic discovery feature giving network engineers end-to-end network topology visibility; mapped 689 peer links by concurrently parsing CDP/LLDP neighbor tables from Cisco NX-OS switches over SSH
- Implemented Pydantic validation layer enforcing typed schemas and sensor bounds at repository boundaries; built AI-automated visual validation of line graphs to catch upstream data bugs in generated charts
- Built reproducible scikit-learn pipeline evaluating peer telemetry for next-day optical forecasts; leakage-resistant splits and a 2K-iteration block bootstrap over 1,284 samples rejected the feature (95% CI crossed zero)

RoboVision Lab | varunpal.info/research

May 2024 – Aug 2025

Summer Researcher

University of Illinois Urbana-Champaign, IL

- Researched a generalizable robotic pick and place task diffusion policy under the guidance of Prof Saurabh Gupta
- Trained neural net policy utilizing self and cross attention on DROID data set in a transformer architecture
- Implemented a data-distributed parallel system, improving GPU utilization from 30% to 70% avg
- Developed system for filtering and processing large video datasets (1.7 TB) for training robotic ML policies

Head TA for ECE 391 (Operating Systems)

Aug 2026 – Present

Head Teaching Assistant

University of Illinois Urbana-Champaign, IL

- Head TA for 300-student OS course; direct 5 TAs and 30 CAs across exam design, MP releases, and grading
- Developed core Unix-like OS components (Shell, Pipes) for student projects; hold 6 weekly office hours

Abbott

May – Sep 2023

Software Engineering Intern

Santa Clara, CA

- Designed an efficient data parsing algorithm using chunking techniques to process large text data (> 4GB) from glucose monitoring devices, enabling faster analysis of product performance metrics
- Created a data visualization dashboard using Power BI, providing real-time insights into product testing results

PROJECT EXPERIENCE

Out-of-Order RISC-V CPU (ECE 411) | SystemVerilog, Computer Architecture, Synopsys

Spring 2026

- Built 7-stage out-of-order RV32IM CPU with explicit register renaming; synthesized at 645 MHz / ~200K area
- Built LLM waveform debugger after finding Codex's RTL-only stall diagnoses unreliable; exposed VCD search tools over MCP so every hypothesis was verified against cycle-level waveform evidence
- Implemented 2-way superscalar front-end through commit and a split LSQ with one-hot age-tagged out-of-order load issue between older stores — +8.3% geo-mean IPC across 6 benchmarks
- Built gshare branch predictor with 64-entry BTB achieving 81.3% branch accuracy (vs. 47.5% baseline)

OSTrich | Operating System, RISC-V, Embedded Systems

Fall 2024

- Built a riscv OS from scratch and placed 3rd out of 100 teams in OS design competition
- Implemented Unix-based OS with paging virtual memory, multi-thread processes, and file system on RISC-V
- Developed UART and VGA drivers, interrupt handling and real-time scheduling

CUDA CNN Convolution Kernel Optimization | CUDA C/C++

Spring 2025

- Optimized convolution kernel for LeNet-5 inference on Fashion MNIST dataset, implementing matrix unrolling, tiled matrix multiplication, and kernel fusion to achieve sub50ms run time on 10,000 images on A100 GPUs
- Optimizations include CUDA Streams, Tensor Cores with Warp Matrix Functions, and FP16 arithmetic

Distributed Transaction System | Go, TCP Networking, Concurrency Control

Spring 2025

- Developed Raft consensus protocol for leader election and log replication, achieving sub-5 second leader recovery times and maintaining linearizability guarantees across distributed replicas with automated failover capabilities
- Designed ACID-compliant transaction manager using timestamp ordering concurrency control, supporting up to 5 concurrent clients with deadlock-free operation through message-passing architecture and two-phase commit

FPGA: Plants vs Zombies Video Game | System Verilog, C, Vivado

Spring 2024

- Recreated PvZ on Spartan 7 FPGA using System Verilog and C with tools like DebugCore and Simulation
- Utilized System on Chip design with MicroBlaze CPU for peripheral control

TECHNICAL SKILLS

Languages: Python, Go, C, C++, CUDA C/C++, SystemVerilog

Systems & Frameworks: MongoDB, Docker, Redis, Linux, Git, PyTorch (DDP), Performance Profiling

Tools: Plotly Dash, Synopsys Design Compiler (Synthesis), VCS (Simulation), Verdi, Vivado/Vitis, CUDA (Nsight, Tensor Cores)

AI-Assisted Development: Codex, Claude Code, MCP tool servers, LLM agent tooling